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Course: Calculus Beta

Assignment: MyLab Math - Uge 1

Given $y = f(u)$ and $u = g(x)$, find $\frac{dy}{dx} = f'(g(x))g'(x)$.

$$y = 9u - 2, \quad u = \left(\frac{1}{3}\right)x^6$$

To find $\frac{dy}{dx}$, use the Chain Rule, which states if $f(u)$ is differentiable at the point $u = g(x)$ and $g(x)$ is differentiable at x , then the composite function $(f \circ g)(x) = f(g(x))$ is differentiable at x , and $(f \circ g)'(x) = f'(g(x)) \cdot g'(x)$.

In Leibniz's notation, if $y = f(u)$ and $u = g(x)$, then $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$, where $\frac{dy}{du}$ is evaluated at $u = g(x)$.

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dx} \\ &= \frac{d}{du}(9u - 2) \cdot \frac{d}{dx}\left(\frac{1}{3}x^6\right)\end{aligned}$$

Use the rules of differentiation to find $\frac{d}{du}(9u - 2)$ and $\frac{d}{dx}\left(\frac{1}{3}x^6\right)$.

$$\frac{d}{du}(9u - 2) = 9, \quad \frac{d}{dx}\left(\frac{1}{3}x^6\right) = 2x^5$$

Simplify.

$$\begin{aligned}&= \frac{d}{du}(9u - 2) \cdot \frac{d}{dx}\left(\frac{1}{3}x^6\right)x^6 \\ &= 9 \cdot 2x^5 \\ &= 18x^5\end{aligned}$$

Thus, $\frac{dy}{dx} = 18x^5$.